

Verified Notes on the Dual Weight Distribution of the Code in Theorem 4

ShuYu

Abstract

Starting from the source Theorem 4 and Table 1, we revisit the dual code of the binary minimal linear code $C_{f,g}$. We first reconcile the parameters and note that the table yields a code with seven nonzero weights. Using the classical MacWilliams identity as input, we then give the exact MacWilliams–Krawtchouk formula for the full dual weight distribution, prove that $A_1^\perp = A_2^\perp = 0$, derive the explicit formulas for $A_3^\perp$, $A_4^\perp$, and $A_5^\perp$, prove that $A_3^\perp > 0$ under the admissible parameter range, and hence obtain $d(C_{f,g}^\perp) = 3$. Finally, we show that if s is even, then the dual weight distribution is symmetric. All new code-specific results proved in this note are formally verified in Lean 4. Only the general MacWilliams identity, used as a standard cited theorem, and the final qualitative remark about $C^\perp$ not being a few-weight code remain outside the current Lean formalization.

1 Source Statement

We begin by recording the source statement visible in the supplied image. The definitions of f , g , and the code $C_{f,g}$ come from the preceding Theorem 3 in the source paper.

Theorem 1 (Source Theorem 4). *Let $s = y$ and $z = 1$. If $m \geq 6$ and $2 \leq s \leq 2^{t-1} - 1$, then the code $C_{f,g}$ in Theorem 3 is a $[2^m - 1, m + 2]$ binary minimal linear code, and its weight distribution is given by Table 1. Furthermore, if $s \leq 2^{t-2}$, then*

$$\frac{w_{\min}}{w_{\max}} \leq \frac{1}{2}. \quad (1)$$

2 Parameters from Source Theorem 4 and Table 1

Let $q = 2^t$. Table 1 gives the weights and multiplicities of the code $C = C_{f,g}$:

Table 1: the weight distribution of $C_{f,g}$ in Theorem 4.

w	A_w
0	1
$s(q - 1)$	2
$(2s - 2)(q - 1)$	1
2^{m-1}	$2^m - 1$
$2^{m-1} - s$	$2(q + 1 - s)(q - 1)$
$2^{m-1} - (2s - 2)$	$(q + 3 - 2s)(q - 1)$
$2^{m-1} + q - s$	$2s(q - 1)$
$2^{m-1} + q - (2s - 2)$	$(2s - 2)(q - 1)$

Thus there are seven nonzero weights, or eight weights in total when the zero weight is included.

The total number of codewords obtained from the table is

$$\sum_w A_w = 1+2+1+(2^m-1)+2(q+1-s)(q-1)+(q+3-2s)(q-1)+2s(q-1)+(2s-2)(q-1) = 2^m+3q^2. \quad (2)$$

Since C is a binary linear code, $|C|$ must be a power of 2. Therefore the integer

$$2^m + 3q^2 = 2^m + 3 \cdot 2^{2t} \quad (3)$$

must be a power of 2.

Proposition 2. *The parameters satisfy*

$$m = 2t. \quad (4)$$

Consequently,

$$|C| = 2^m + 3q^2 = 4q^2 = 2^{m+2}. \quad (5)$$

Proof. If $m < 2t$, then

$$|C| = 2^m(1 + 3 \cdot 2^{2t-m}), \quad (6)$$

and the factor $1 + 3 \cdot 2^{2t-m}$ is an odd integer strictly larger than 1, so $|C|$ cannot be a power of 2.

If $m > 2t$, then

$$|C| = 2^{2t}(2^{m-2t} + 3), \quad (7)$$

and the factor $2^{m-2t} + 3$ is again an odd integer strictly larger than 1, so $|C|$ cannot be a power of 2.

Hence neither $m < 2t$ nor $m > 2t$ is possible, and therefore $m = 2t$. Substituting this into $|C| = 2^m + 3q^2$ gives $|C| = 4q^2 = 2^{m+2}$. $\square$

This parameter reconciliation is formally verified in Lean 4.

After this derivation, the weights may be rewritten as

$$2^{m-1} = q^2/2, \quad (8)$$

so the nonzero weights become

$$s(q-1), (2s-2)(q-1), q^2/2, q^2/2-s, q^2/2-(2s-2), q^2/2+q-s, q^2/2+q-(2s-2), \quad (9)$$

and the length is

$$n = 2^m - 1 = q^2 - 1. \quad (10)$$

3 The dual weight enumerator

Let

$$W_C(x) = \sum_{i=0}^n A_i x^i \quad (11)$$

be the weight enumerator of C , and let

$$W_{C^\perp}(x) = \sum_{j=0}^n A_j^\perp x^j \quad (12)$$

be the weight enumerator of the dual code.

Theorem 3. For $0 \leq j \leq n = q^2 - 1$,

$$A_j^\perp = \frac{1}{4q^2} \sum_w A_w K_j(w), \quad (13)$$

where the sum runs over the eight weights listed above and

$$K_j(w) = \sum_{\ell=0}^j (-1)^\ell \binom{w}{\ell} \binom{n-w}{j-\ell} \quad (14)$$

is the binary Krawtchouk polynomial.

Equivalently,

$$W_{C^\perp}(x) = \frac{(1+x)^n}{4q^2} W_C\left(\frac{1-x}{1+x}\right). \quad (15)$$

Proof. Since $|C| = 4q^2$, the binary MacWilliams identity gives

$$W_{C^\perp}(x) = \frac{(1+x)^n}{|C|} W_C\left(\frac{1-x}{1+x}\right) = \frac{(1+x)^n}{4q^2} W_C\left(\frac{1-x}{1+x}\right). \quad (16)$$

Extracting the coefficient of x^j yields the Krawtchouk expansion. $\square$

Remark 4. The theorem above is used as a classical coding-theoretic input and is cited rather than formalized here. What is formally verified in Lean 4 are the new code-specific consequences extracted from this formula in the rest of the note.

4 The first dual coefficients

Let

$$M_r = \sum_w A_w w^r \quad (17)$$

denote the r -th power moment of C .

Lemma 5. The first two power moments are

$$M_1 = 2q^2(q^2 - 1), \quad M_2 = q^4(q^2 - 1). \quad (18)$$

Proof. This follows by direct substitution of the eight entries from Table 1 and straightforward simplification. $\square$

These two power-moment identities are formally verified in Lean 4.

Proposition 6. The dual code satisfies

$$A_1^\perp = 0, \quad A_2^\perp = 0. \quad (19)$$

Proof. Since $|C| = 4q^2 = 2^{m+2}$, the dimension is $k = m + 2$, and $n = q^2 - 1$.

By the first two Pless power moments,

$$M_1 = 2^{k-1}(n - A_1^\perp), \quad (20)$$

and

$$M_2 = 2^{k-2}(n(n+1) - 2nA_1^\perp + 2A_2^\perp). \quad (21)$$

Using $2^k = 4q^2$, $n = q^2 - 1$, $M_1 = 2q^2(q^2 - 1)$, and $M_2 = q^4(q^2 - 1)$, the first identity yields $A_1^\perp = 0$, and then the second yields $A_2^\perp = 0$. $\square$

The vanishing statements $A_1^\perp = 0$ and $A_2^\perp = 0$ are formally verified in Lean 4.

5 Explicit low-weight multiplicities

Theorem 7. *The number of codewords of weight 3 in $C^\perp$ is*

$$A_3^\perp = \frac{1}{6} \left(q^4 + (3 - 6s)q^3 + (18s^2 - 26s + 10)q^2 + (-20s^3 + 48s^2 - 40s + 12)q + (20s^3 - 66s^2 + 72s - 26) \right). \quad (22)$$

Proof. From the third Pless power moment and the already proved equalities $A_1^\perp = A_2^\perp = 0$, one obtains

$$M_3 = 2^{k-3} (n^2(n+3) - 6A_3^\perp). \quad (23)$$

Substituting the Table 1 weights and multiplicities into M_3 and simplifying gives the stated formula. $\square$

This formula is formally verified in Lean 4. In the formal development, it is also checked independently from the Krawtchouk-polynomial expression.

Proposition 8. *For the admissible range $q = 2^t$, $t \geq 3$, and $2 \leq s \leq q/2 - 1$, one has*

$$A_3^\perp > 0. \quad (24)$$

Hence

$$d(C^\perp) = 3. \quad (25)$$

Proof. Set

$$r = q - 2s. \quad (26)$$

Then $2 \leq r \leq q - 4$. Rewriting the formula for $A_3^\perp$ gives

$$A_3^\perp = \frac{q-1}{12} F(q, r), \quad (27)$$

where

$$F(q, r) = 3q^2r - 6qr^2 + 5r^3 + 9q^2 - 34qr + 33r^2 - 44q + 72r + 52. \quad (28)$$

For fixed r , we have

$$\frac{\partial F}{\partial q} = 6qr - 6r^2 + 18q - 34r - 44. \quad (29)$$

Since $q \geq r + 4$, this implies

$$\frac{\partial F}{\partial q} \geq 6r(r+4) - 6r^2 + 18(r+4) - 34r - 44 = 8r + 28 > 0. \quad (30)$$

Thus $F(q, r)$ is increasing in q , and so

$$F(q, r) \geq F(r+4, r) = 2(r^3 + 4r^2 + 6r + 10) > 0. \quad (31)$$

Therefore $A_3^\perp > 0$. Together with $A_1^\perp = A_2^\perp = 0$, this proves $d(C^\perp) = 3$. $\square$

The positivity statement $A_3^\perp > 0$ and the conclusion $d(C^\perp) = 3$ are formally verified in Lean 4.

Proposition 9. *The next two coefficients are*

$$A_4^\perp = \frac{1}{24} \left(q^6 - 8q^5s + 36q^4s^2 - 80q^3s^3 + 72q^2s^4 + 4q^5 - 40q^4s + 120q^3s^2 \right. \\ \left. - 16q^2s^3 - 288qs^4 + 13q^4 - 56q^3s - 324q^2s^2 + 1104qs^3 + 216s^4 \right. \\ \left. - 4q^3 + 488q^2s - 1632qs^2 - 1008s^3 - 198q^2 + 1056qs + 1800s^2 \right. \\ \left. - 240q - 1440s + 424 \right), \quad (32)$$

and

$$A_5^\perp = \frac{1}{120} \left(q^8 - 10q^7s + 60q^6s^2 - 200q^5s^3 + 360q^4s^4 - 272q^3s^5 + 5q^7 - 70q^6s \right. \\ \left. + 360q^5s^2 - 680q^4s^3 - 160q^3s^4 + 1360q^2s^5 + 20q^6 - 220q^5s + 180q^4s^2 \right. \\ \left. + 2560q^3s^3 - 4600q^2s^4 - 2720qs^5 + 30q^5 + 508q^4s - 4520q^3s^2 + 4400q^2s^3 \right. \\ \left. + 14240qs^4 + 1632s^5 - 264q^4 + 2672q^3s + 2560q^2s^2 - 30720qs^3 - 9840s^4 \right. \\ \left. - 440q^3 - 6080q^2s + 32800qs^2 + 24640s^3 + 2440q^2 - 16960qs - 31440s^2 \right. \\ \left. + 3360q + 20160s - 5152 \right). \quad (33)$$

Proof. These formulas follow by direct evaluation of the MacWilliams–Krawtchouk expression for $A_j^\perp$ with $j = 4, 5$. Specifically, using the Krawtchouk polynomials in the variable $D = n - 2w$,

$$24K_4(w) = D^4 - (6n - 8)D^2 + 3n(n - 2), \quad (34)$$

$$120K_5(w) = D^5 - (10n - 20)D^3 + (15n^2 - 50n + 24)D, \quad (35)$$

we substitute the D -values for each table weight and verify that

$$\sum_w A_w D_w^4 = 4q^2 (\text{numerator of } 24A_4^\perp + n(3n - 2)), \quad (36)$$

$$\sum_w A_w D_w^5 = 4q^2 (\text{numerator of } 120A_5^\perp + (10q^2 - 30) \cdot \text{numerator of } 6A_3^\perp). \quad (37)$$

Both identities are polynomial equalities in q and s , verified formally in Lean by the `ring` tactic. $\square$

6 Symmetry for even s

Proposition 10. *If s is even, then*

$$A_j^\perp = A_{n-j}^\perp \quad (0 \leq j \leq n). \quad (38)$$

Proof. When s is even, every weight appearing in Table 1 is even: $q^2/2$ is even because $q = 2^t$ with $t \geq 3$, and the remaining seven weights are visibly even as well. Hence every codeword of C has even Hamming weight, so C is an even code. Therefore the all-ones vector $\mathbf{1}$ lies in $C^\perp$. The map

$$c \mapsto c + \mathbf{1} \quad (39)$$

is then a bijection on $C^\perp$ sending words of weight j to words of weight $n - j$, and the symmetry follows. $\square$

This symmetry statement is formally verified in Lean 4.

Remark 11. *The MacWilliams formula gives the complete dual weight distribution. In general, $C^\perp$ is not a few-weight code, so the coefficients $A_j^\perp$ are most naturally described by the Krawtchouk expansion rather than by a short finite list. This is a qualitative observation rather than a formalized theorem in the current Lean development.*